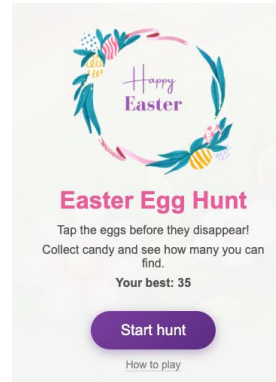


Data vs Statistics

Module 4 - Descriptive Analytics & Visualization

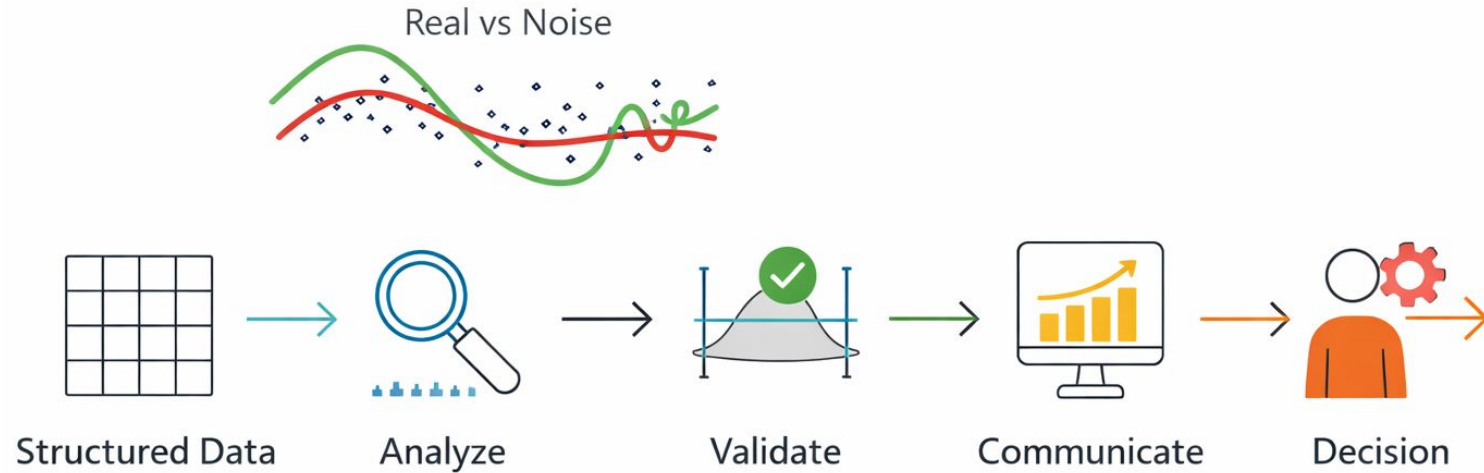
Get to know Lana — Fun Fact



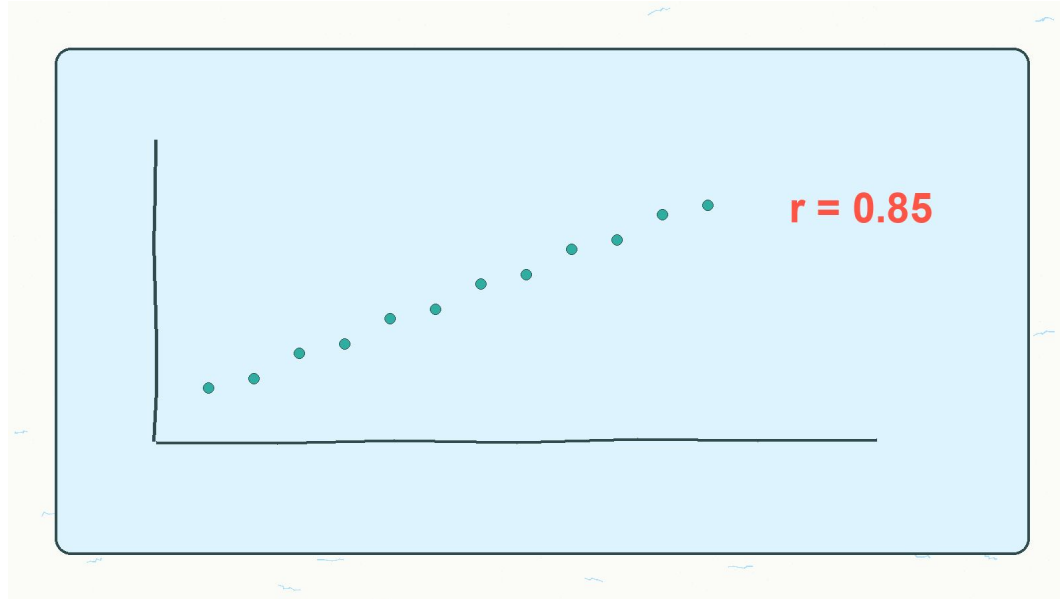
Celebrating Easter today!

- Easter is the most important holiday in the Christian faith
- Worship service, celebratory music, family meal – often Easter brunch, big party with special foods
- A few years ago I made a webapp Easter egg hunt game for fun
 - Play it: easter-egg-basket.vercel.app

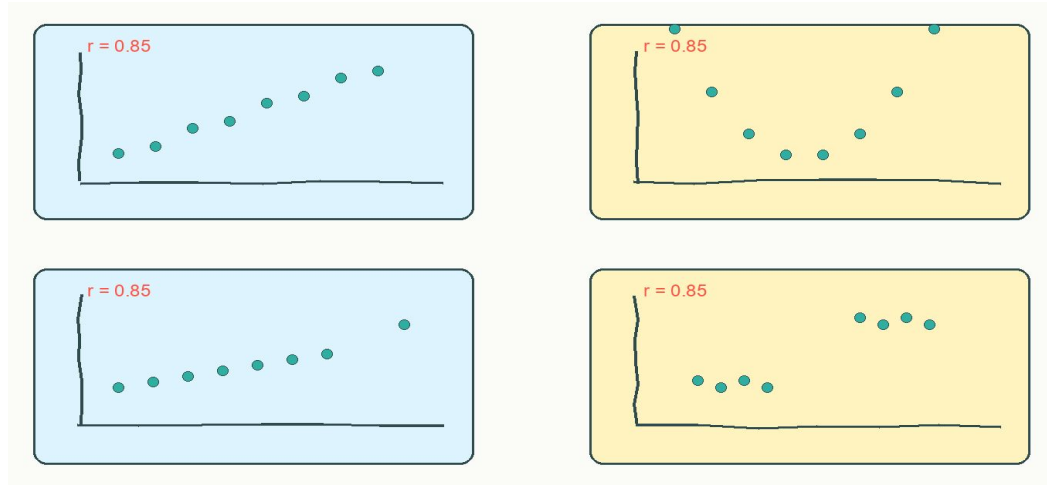
The Journey: M3 to M4



Would You Invest?

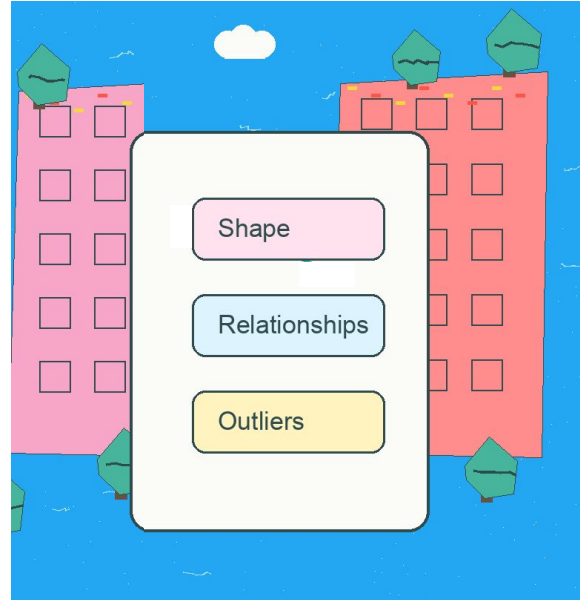


Same Number



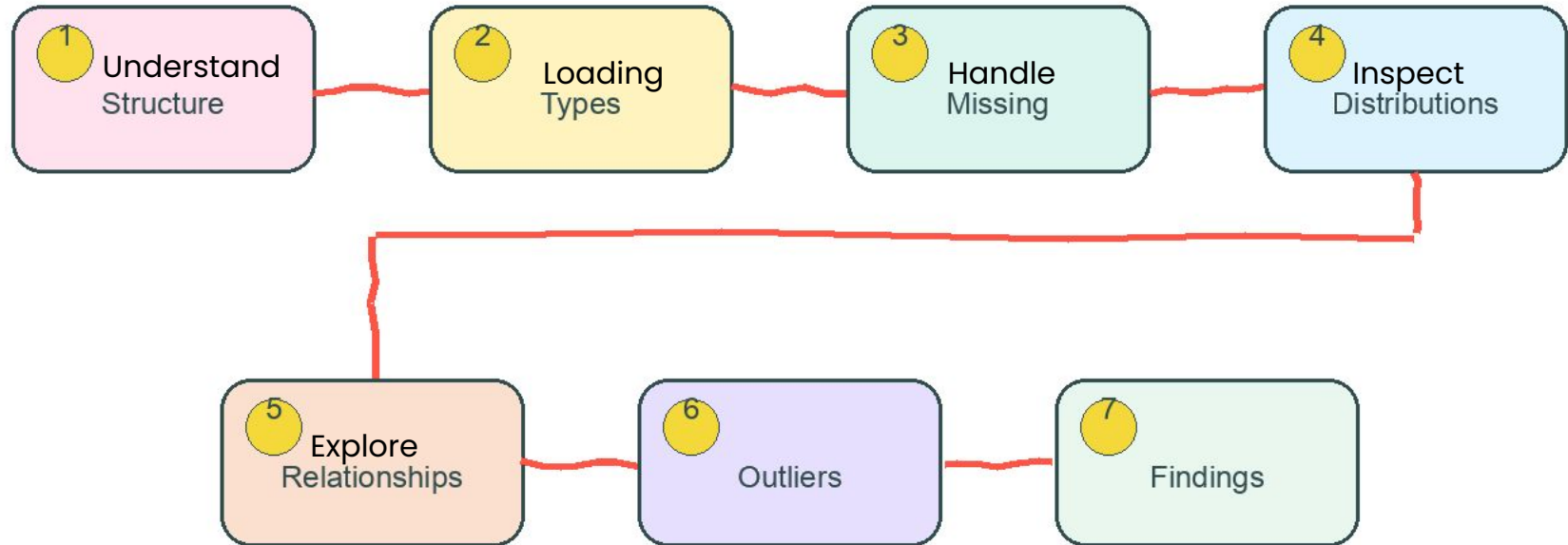
- Same r value
- Different patterns
- Always inspect the plot

Look First: Exploratory Data Analysis (EDA)

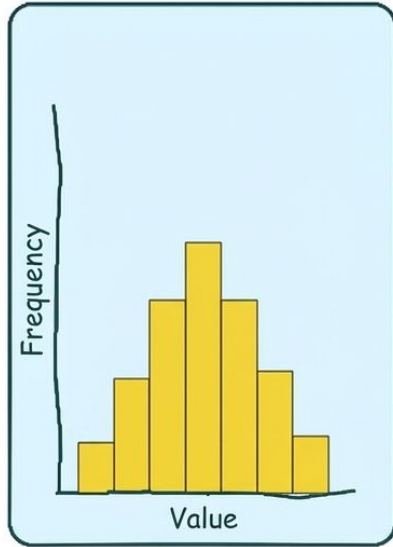


- Shape: what is normal?
- Relationships: what moves together?
- Outliers: what breaks the pattern?

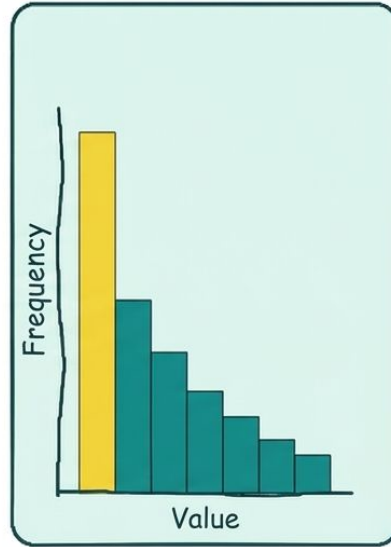
EDA Workflow



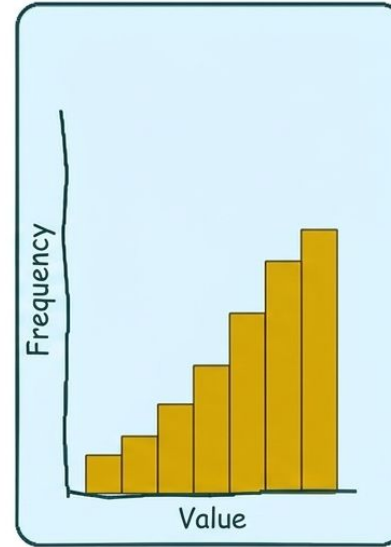
Distribution Shape



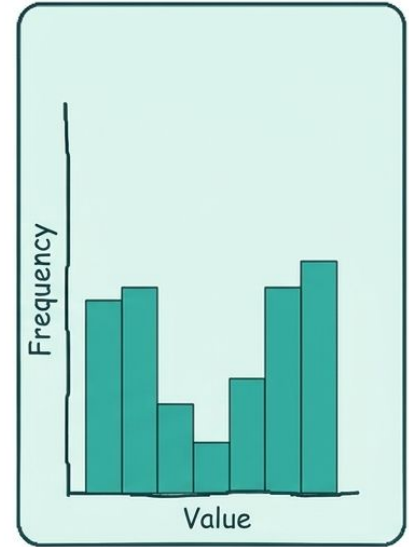
Symmetric



Right Skewed

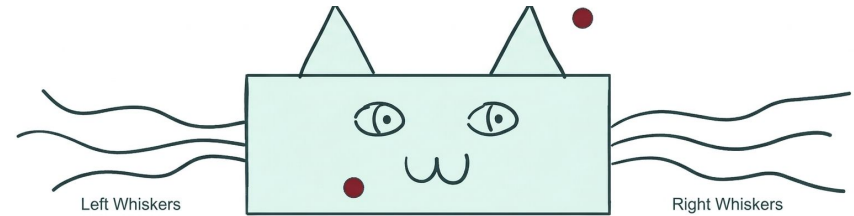
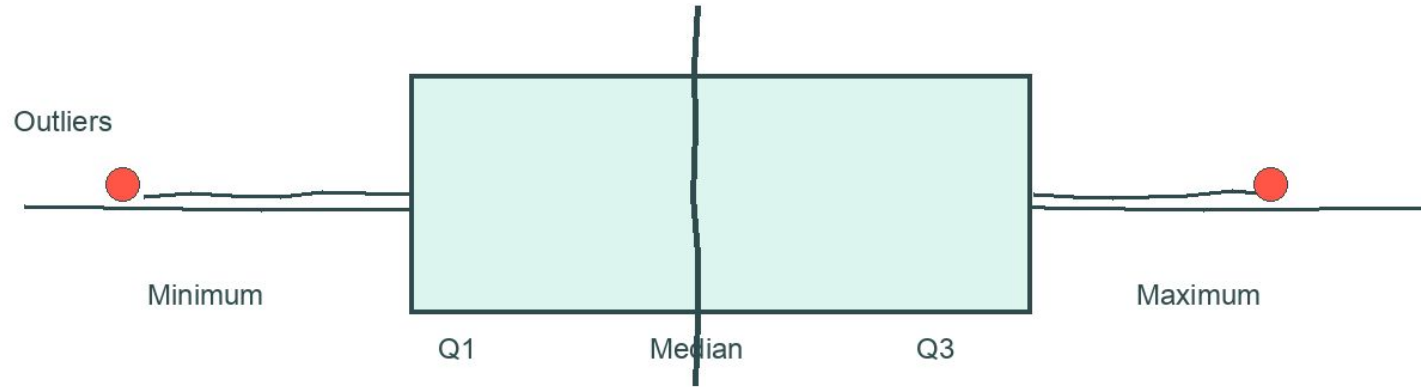


Left Skewed

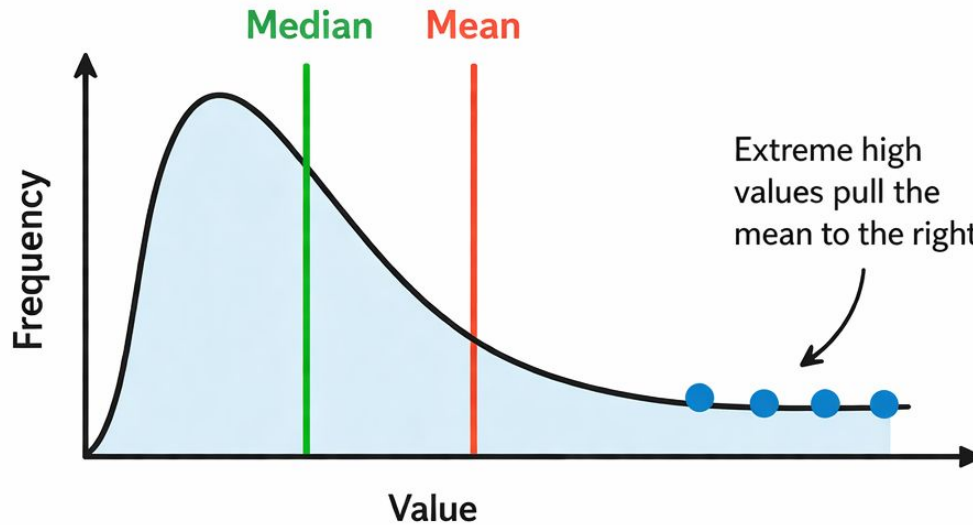


Bimodal

How to Read a Box Plot



Right-Skewed Data: Mean or Median?



Use the MEDIAN.

It represents the typical observation and isn't affected by extreme values.



The MEAN gets pulled toward the long right tail.

A few extreme values can inflate it and make it misleading.

Amman Delivery



Inspect First

1 `df.shape`

Orders: 2,000
Columns: 12

how big is the dataset?

2 `df.dtypes`

delivery_time numeric
cuisine categorical
order_date datetime

what types are the columns?

3 `df.isnull().sum()`

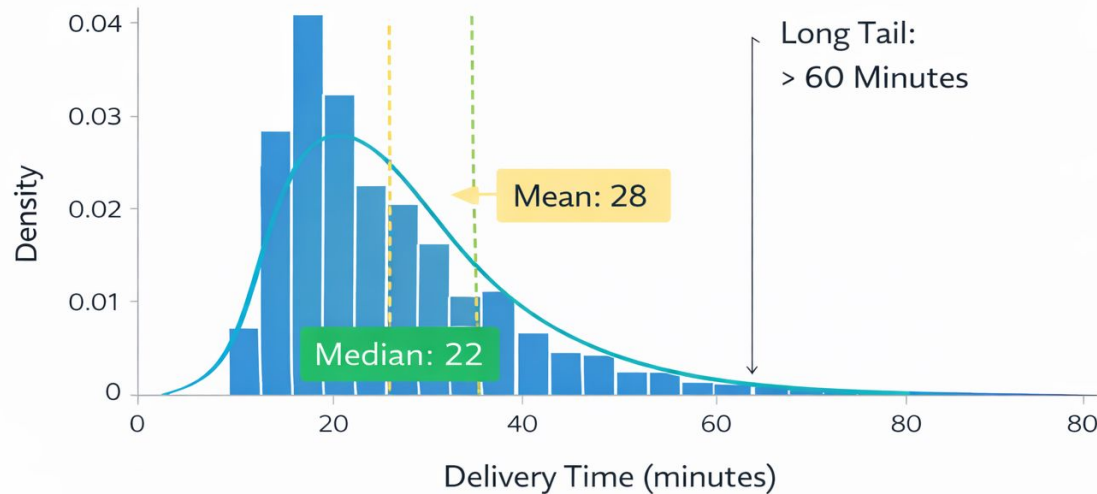
restaurant_rating	150
- ratings left empty (2,000 orders)	7%

where is data missing?

Reading the Distribution

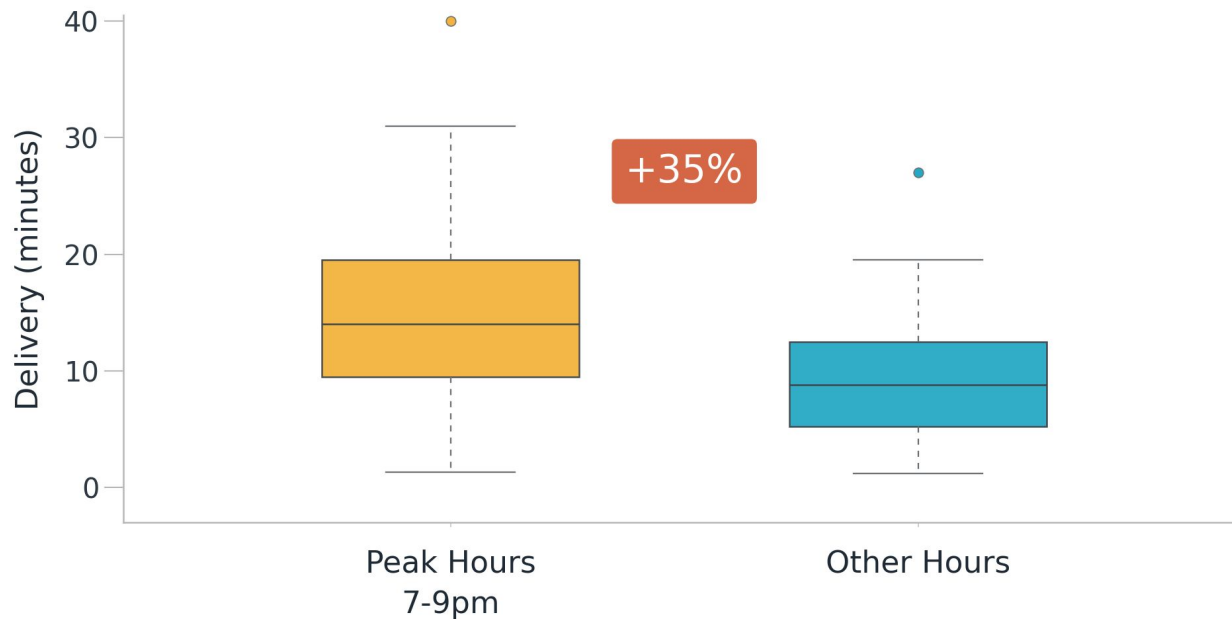
Delivery Times: Right-Skewed

- Most deliveries: 15-30 minutes
- Long tail: some take over an hour

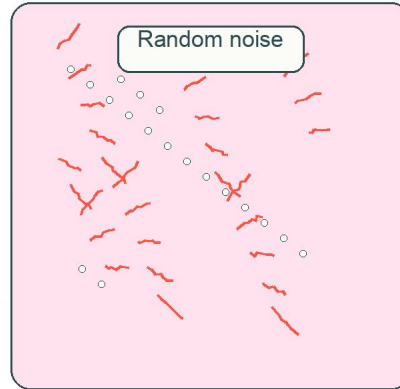
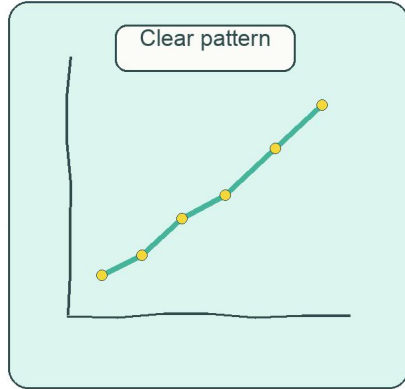


The Annotated Finding

Peak-Hour Deliveries Take 35 Percent Longer

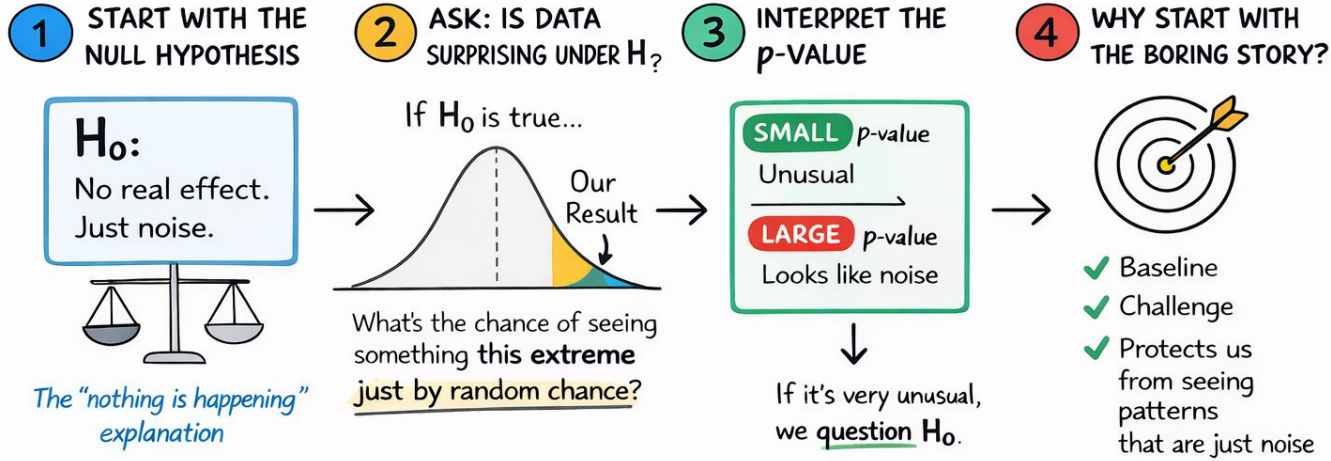


From Pattern to Significance: Inference

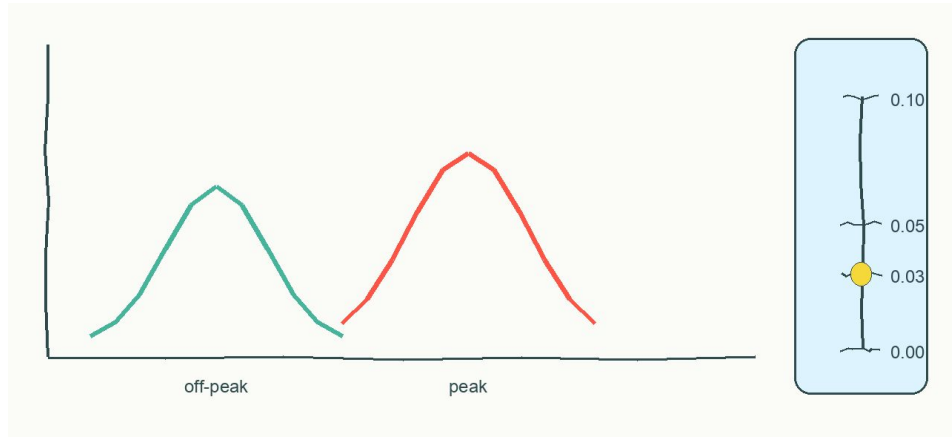


- Start with a visible pattern
- Ask whether chance could explain it
- Use testing to separate signal from noise

Hypothesis Testing Logic

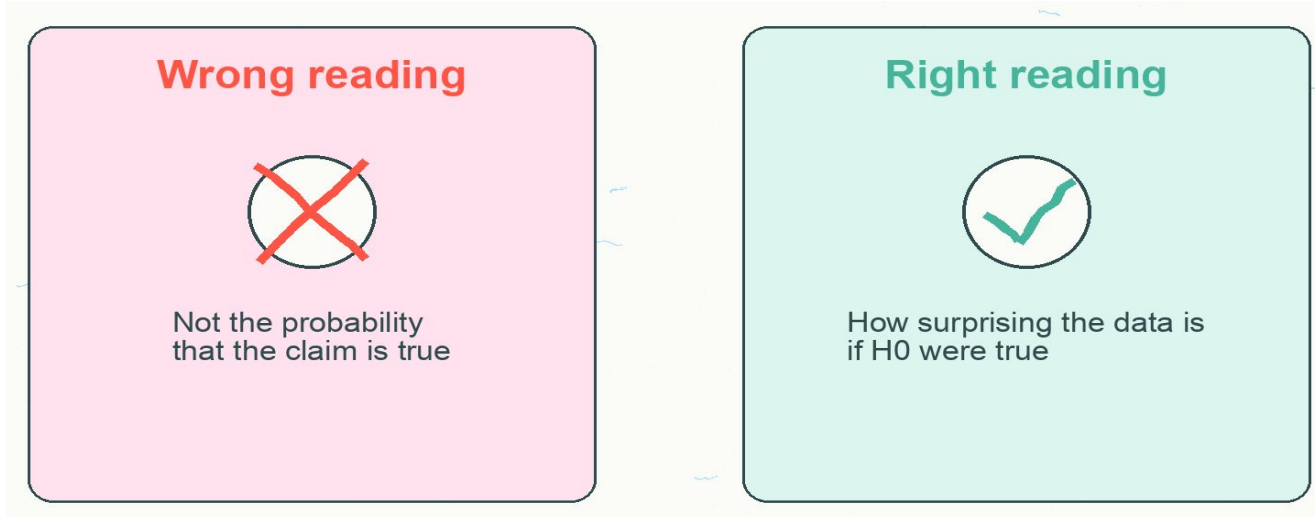


Interpreting $p = 0.03$



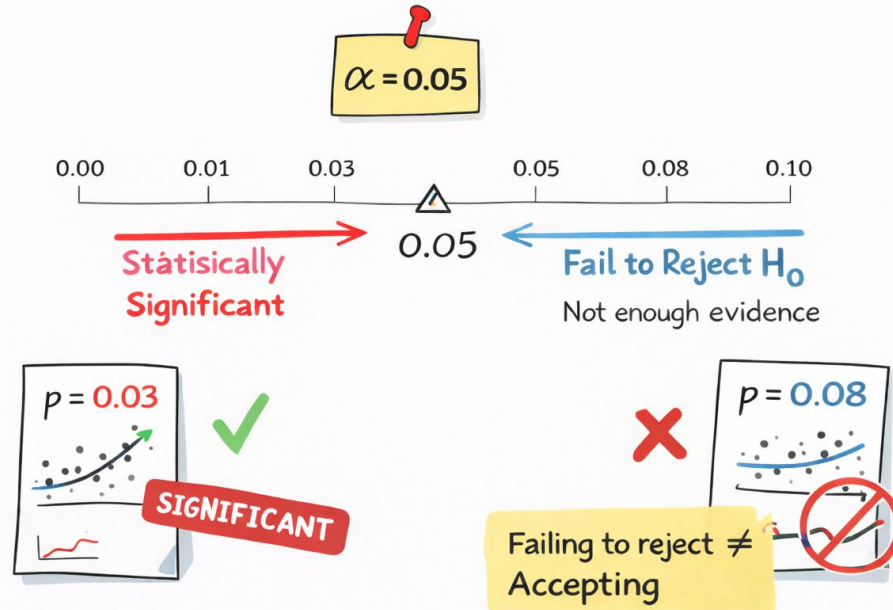
- Compare peak vs off-peak
- Result: $p = 0.03$
- This gap would be rare by chance

What a p-Value Means

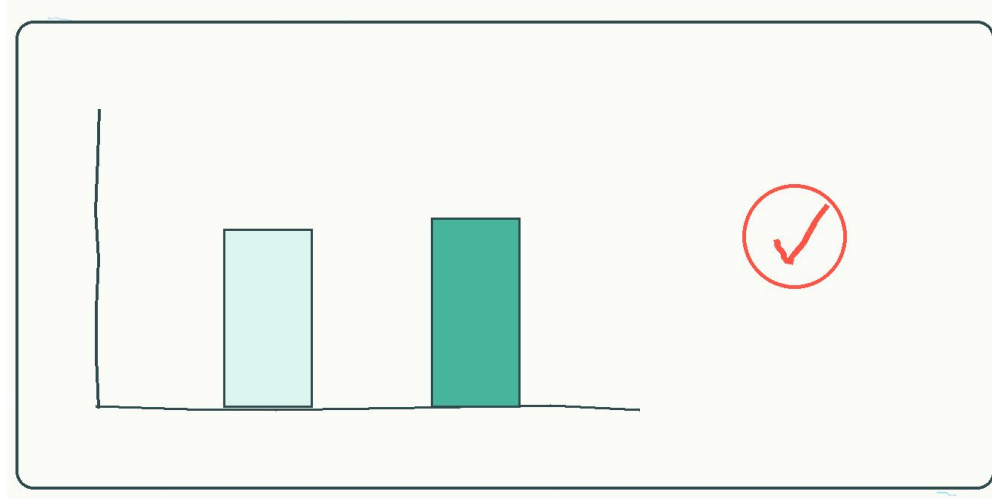


The Significance Threshold

How Small Does a p-value Need to Be?



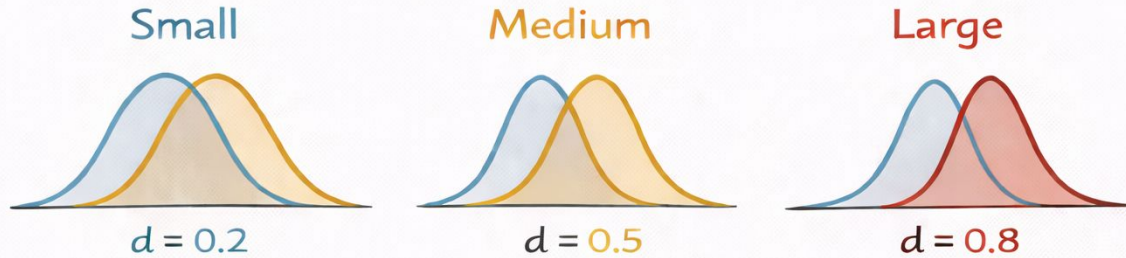
Statistical vs Practical Significance



- Significant = unlikely by chance
- Important = large enough to matter
- Report both

Effect Size

Difference between means in standard deviations

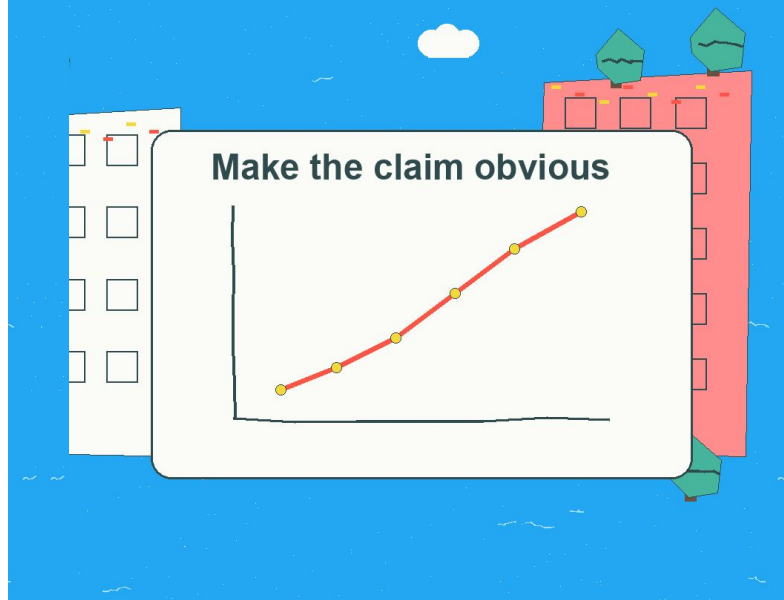


t-test
 $p = 0.0001$ ✓

p-value: Is the difference real?

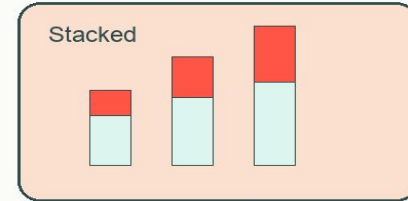
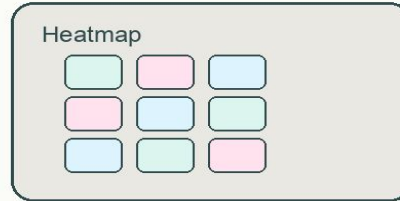
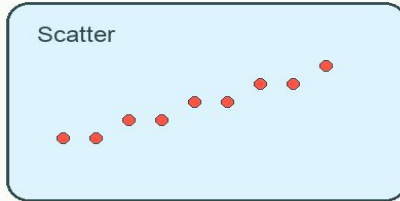
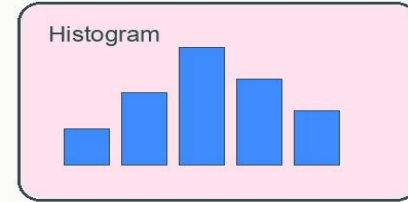
effect size: Is the difference
big enough to care about?

A Chart Is an Argument



- Make one clear claim
- Titles should state the finding
- Reduce decoding work

Choose the Chart



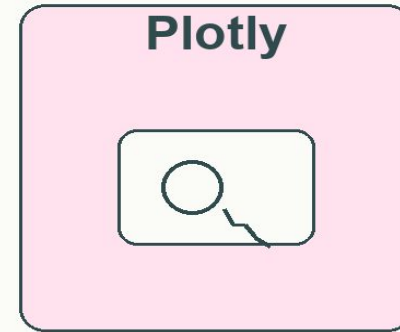
Picking Tools



fast + statistical

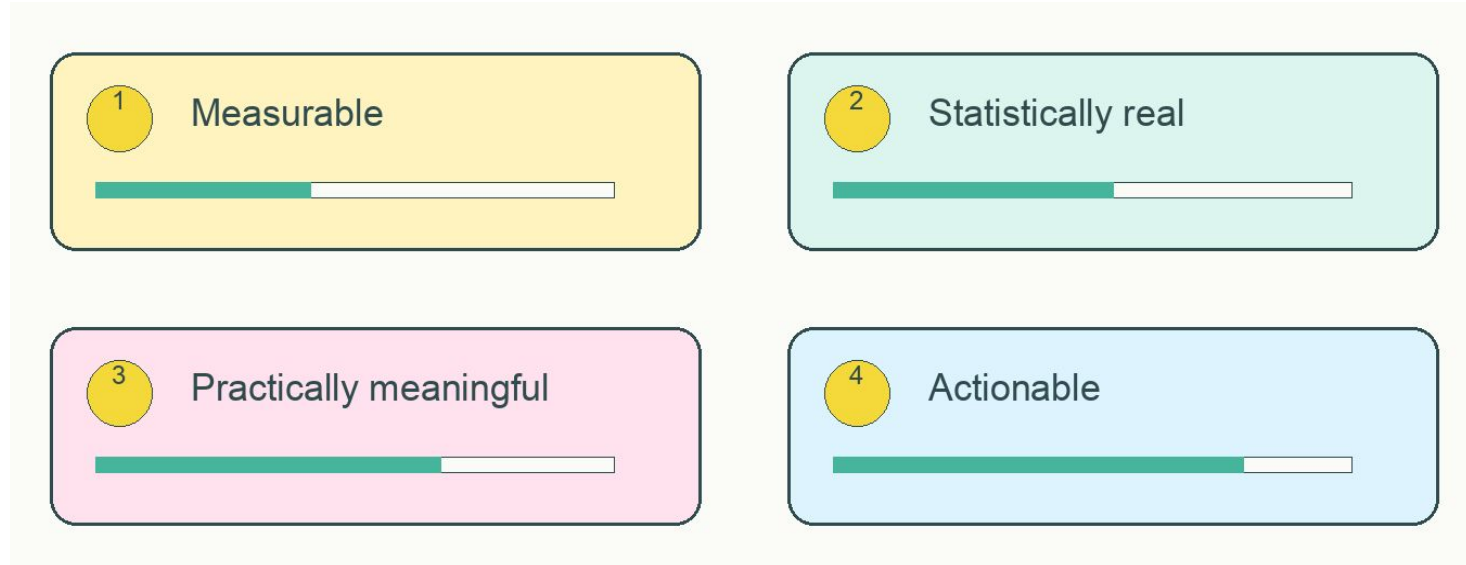


precise + custom



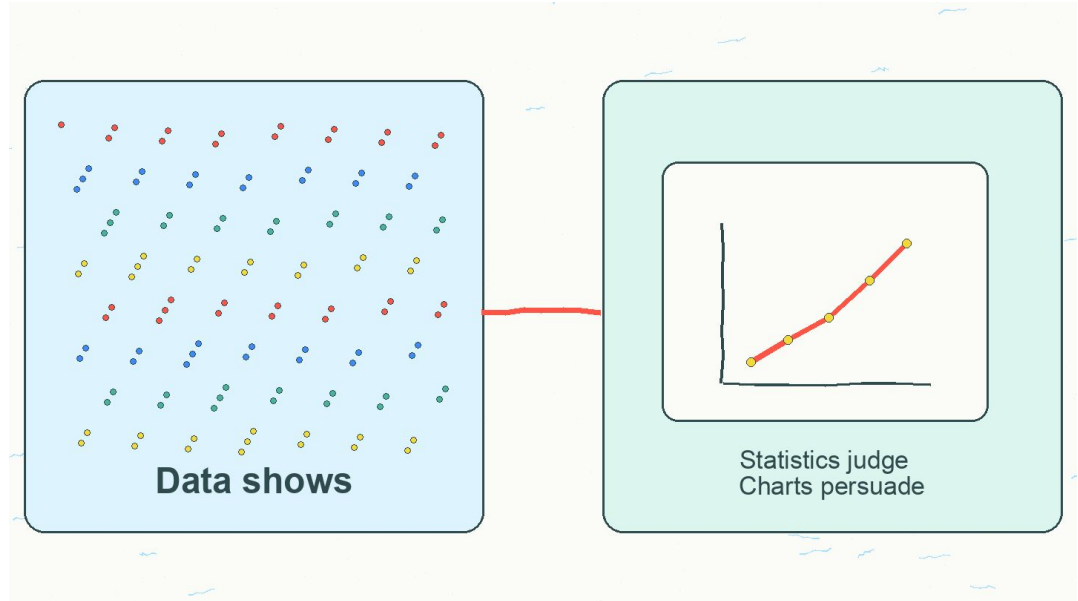
interactive

What Makes a Good KPI?

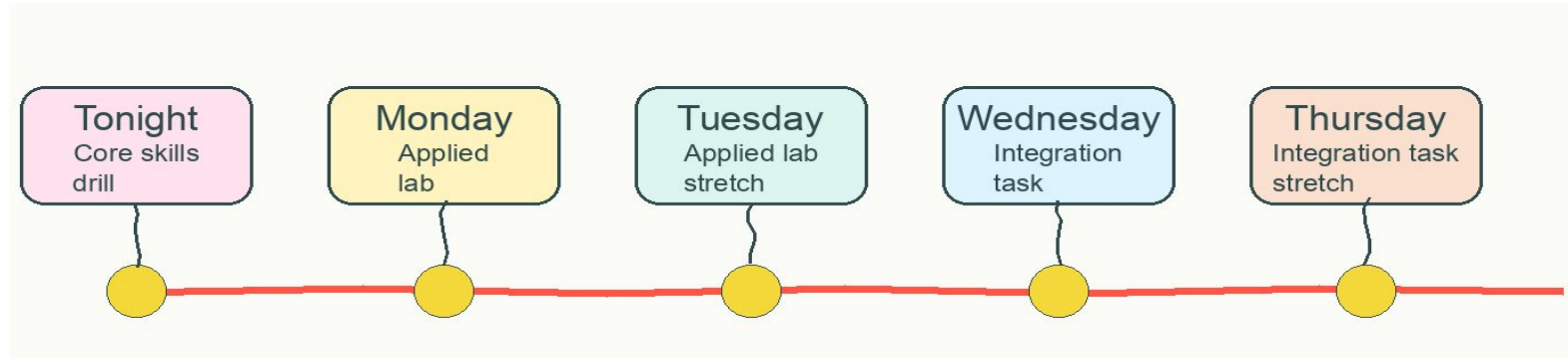


Module Summary

- Data shows what happened
- Statistics judge whether it matters
- Charts persuade people to act



What Happens Next



Questions?

Announcements

Congratulations!



Cohort Mentors

- Zaid N.
- Hadeel

Module Mentors

- Shahd
- Nayef